Formula Ceiling Detection: Phased Enumeration for Interpretable Symbolic Prediction

Andrew H. Bond
Department of Computer Engineering, San Jose State University
Senior Member, IEEE ORCID: 0009-0003-2599-6158
andrew.bond@sjsu.edu

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Abstract

We introduce a framework for discovering interpretable predictive formulas from labeled scientific data by enumerating compositions of mathematical operations in phases of increasing complexity. Given a feature matrix and binary labels, the method evaluates all single features, all pairwise operation combinations, and selected higher-order compositions, finding the formula that maximizes F1 score within each phase. GPU-vectorized evaluation enables thousands of candidates per second.

The framework's primary contribution is the *formula ceiling*: the highest F1 achieved by any enumerated formula. Comparing this ceiling to a black-box ensemble (trained and evaluated on the same data split) yields the *complexity gap*—how much predictive structure lies beyond the enumerated formula space.

Applied to predicting periodic orbits in the gravitational three-body problem (15,000 configurations, 23 features, 5-fold cross-validated), the method discovers that periodicity is predicted by $\sqrt{n_{\text{clusters}}^2 + \lambda_{\min}^2}$ (ceiling F1 = 0.48). A gradient-boosted ensemble on the same data and splits reaches F1 = 0.83, yielding a complexity gap of 0.35. Across five classification benchmarks, the ceiling correctly identifies problems admitting exact formulas (Circles: ceiling = 0.996, beating the ensemble) and problems requiring complex models (three-body: gap = 0.35).

Code and data: `https://github.com/ahb-sjsu/astar-symbolic-discovery`.

1 Introduction

A central goal of computational science is to discover interpretable laws from data. This problem has been addressed by genetic symbolic regression [1, 4], neural equation discovery [2], and classical AI systems [3]. Genetic programming explores large formula spaces but provides no optimality guarantee at a given depth and often produces bloated expressions. Neural methods discover accurate but opaque representations. Classical systems require domain-specific search heuristics.

We propose *phased enumeration*: exhaustive evaluation of all formulas within a series of structural templates of increasing complexity. Phase 1 evaluates all single features; Phase 2 evaluates all ordered pairs of features composed with each binary operation; Phase 3 applies unary transforms to the top Phase 2 results. Within Phases 1–2, every candidate is evaluated, so the best formula *within that phase's template* is found by complete enumeration. Phase 3 is restricted to the top-K Phase 2 results and is therefore not exhaustive over all depth-3 formulas. Formulas with structures not matching any template (e.g., multi-term sums, deeply nested compositions) are not searched.

Our framework's primary contribution is the *formula ceiling*:

1. The **ceiling** at each phase: the highest F1 achieved by any enumerated formula. If the ceiling stabilizes across phases, the search has likely saturated for the given operation set.

2. The **complexity gap**: the difference between a black-box ensemble's F1 and the ceiling, both evaluated on the same data using the same cross-validation splits. A positive gap indicates predictive structure beyond the enumerated formula space; a zero or negative gap indicates a simple formula suffices.

2 Method

2.1 Problem Formulation

Definition 1 (Symbolic Prediction Problem)**.** *Given a feature matrix* $X \in \mathbb{R}^{N \times d}$ *and binary labels* $y \in \{0,1\}^N$*, find a formula* $f : \mathbb{R}^d \to \mathbb{R}$ *and threshold* $\tau \in \mathbb{R}$ *that maximize* $F_1(\hat{y}, y)$ *where* $\hat{y}_i = \mathbf{1}[f(X_i) > \tau]$*, subject to* $\mathrm{depth}(f) \leq D$*.*

2.2 Operation Registry

The search space is defined by three operation classes:

Leaf nodes (depth 0): Each of the d features x_j.

Unary operations (depth +1): The full registry contains 8 operations: $\log|\cdot|$, $\sqrt{|\cdot|}$, $(\cdot)^2$, $1/(\cdot)$, $-(\cdot)$, $|\cdot|$, $\sigma(\cdot)$, $\tanh(\cdot)$. In the experiments reported here, we use a subset of 5: $\{\log, \sqrt{}, (\cdot)^2, 1/(\cdot), -(\cdot)\}$.

Binary operations (depth +1): The full registry contains 10 operations. In our experiments we use 7: $\{+, -, \times, \div, \max, \min, \mathrm{hypot}\}$. The remaining 3 (diff-squared, harmonic mean, geometric mean) are available in the software but were not used in the reported runs.

2.3 Phased Exhaustive Search

Rather than a single tree search, we organize the exploration into four phases (Algorithm 1). Phases 1–2 enumerate all formulas of their structural type; Phases 3–4 are restricted to top-K results from earlier phases and are not exhaustive.

Threshold sweep. For each candidate formula f, we compute $v_i = f(X_i)$ for all N samples, then evaluate F1 at 11 percentile thresholds $\{20, 30, 40, \ldots, 95\}$ of the value distribution. Both $v > \tau$ and $v < \tau$ predictors are tested. The best threshold is retained. This discrete sweep approximates but does not guarantee the globally optimal threshold; in practice, F1 is piecewise-constant in τ (changing only when τ crosses a data point), and our 11 percentiles sample this step function densely enough that the reported F1 matches exhaustive threshold search on a verified subset (median absolute error < 0.001). The sweep is vectorized and adds negligible cost.

Cross-validation protocol. To ensure a fair comparison between the formula ceiling and the ensemble baseline, both are evaluated using the same 5-fold stratified split. For each fold: (i) the phased search runs on the training portion (formula selection and threshold tuning); (ii) the best formula + threshold is evaluated on the held-out test portion; (iii) the gradient-boosted ensemble is trained and evaluated on the same train/test split. The reported ceiling and ensemble F1 are both averages over the 5 held-out folds. This prevents information leakage from threshold selection into the ceiling estimate.

GPU acceleration. Features are stored as GPU arrays (CuPy). All formula evaluations and threshold sweeps are vectorized GPU operations. Phase 2 evaluates $d(d-1) \times |\mathcal{B}|$ formulas (e.g., $23 \times 22 \times 7 = 3{,}542$ with our 7-operation set) in seconds.

Completeness within each phase. Phases 1–2 are exhaustive over all feature (pairs) and operations. Phase 3 applies all unary transforms to the top-K results from Phase 2. Phase 4

Algorithm 1 Phased Symbolic Search (Phases 1–2 exhaustive; 3–4 filtered)

Require: Features $X \in \mathbb{R}^{N \times d}$, labels y, operations $\mathcal{U}$ (unary), $\mathcal{B}$ (binary)
Ensure: Best formula and F1 at each depth

Phase 1: Single features (depth 1)
for $j = 1, \ldots, d$ **do**
  Evaluate $F_1^*(x_j)$ via threshold sweep
$R_1 \leftarrow$ top-K features by F1

Phase 2: Pairwise combinations (depth 2)
for $x_i, x_j \in \{1, \ldots, d\}$, $b \in \mathcal{B}$ **do**
  Evaluate $F_1^*(b(x_i, x_j))$ via threshold sweep
$R_2 \leftarrow$ top-K pairwise formulas

Phase 3: Unary transforms (depth 3)
for $f \in R_2$, $u \in \mathcal{U}$ **do**
  Evaluate $F_1^*(u(f))$ via threshold sweep
$R_3 \leftarrow$ top-K depth-3 formulas

Phase 4: Triple combinations (depth 3)
for $x_i, x_j, x_k \in R_1$, pattern p **do**
  Evaluate $F_1^*(p(x_i, x_j, x_k))$ via threshold sweep
return best formula across all phases

tests multiple combination patterns over the top-K single features. The search is complete at each phase for the specified formula structure; no formulas of that structure are missed.

2.4 Scalability

The formula count per phase is: Phase 1: $d(1+|\mathcal{U}|)$; Phase 2: $d(d-1)|\mathcal{B}|$; Phase 3: $K|\mathcal{U}|$; Phase 4: $\binom{K'}{3}P$. Here K is the number of top pairwise results carried forward, K' the top single features, and P the combination patterns.

For $d = 23$, $|\mathcal{B}| = 7$, $|\mathcal{U}| = 5$, $K = 30$, $K' = 8$, $P = 4$, the total is approximately 5,600 formulas—evaluated in under one minute on a single GPU. Composing Phase 2 with itself (all pairwise-of-pairwise formulas) would require $O(d^2(d-1)^2 \cdot |\mathcal{B}|^2) \approx 12$ million formulas, tractable with GPU vectorization but requiring hours. We report results through Phase 3 (depth 3); Phase 4 triples are also depth 3 but with a different structural template.

2.5 Formula Ceiling and Irreducible Complexity Gap

Definition 2 (Formula Ceiling)**.** *The formula ceiling C_D at depth D is the maximum F1 achieved by any formula of depth $\leq D$ found across all phases: $C_D = \max_{phases} F_1^*$.*

Definition 3 (Irreducible Complexity Gap)**.** *Given a black-box model with F1 score F_1^{bb} trained on the same features, the irreducible complexity gap at depth D is $G_D = F_1^{\text{bb}} - C_D$.*

$G_D > 0$ means the data contains predictive structure that was not captured by any depth-D formula within the searched space. Because the search is exhaustive within each phase template but does not cover all possible depth-D formulas, G_D is an upper bound on the true gap—a formula structure not in our template set could in principle achieve higher F1. However, if C_D

is stable across multiple operation sets and phase structures, this suggests (but does not prove) that the gap is intrinsic to the depth constraint rather than a limitation of the search.

Proposition 1. *C_D is non-decreasing in D and bounded above by F_1^{Bayes}. If $C_{D+1} = C_D$ for a sufficiently rich operation set, the ceiling has converged.*

3 Benchmark Validation

To validate the method beyond the three-body case study, we run the phased search on four standard classification datasets and compare the formula ceiling to gradient-boosted ensembles (Table 1).

Table 1: Formula ceiling across domains. Both ceiling and GB use 5-fold CV: formula selection and threshold tuning occur on training folds only, with F1 measured on each held-out fold. Gap = GB − C.

Dataset (d)	Ceiling	GB F1 (CV)	Gap	Formula
Circles (2)	0.996	0.989±.01	−0.01	`hypot`
Breast Ca. (30)	0.948	0.962±.01	+0.01	`f7+f22`
Ionosphere (34)	0.902	0.934±.03	+0.03	`log(f6 min f5)`
Moons (2)	0.833	0.969±.01	+0.14	`x2`
Three-body (23)	0.480	0.830±.02	+0.35	`hypot(n,` λ`)`

On **Circles**, the method discovers the exact analytic decision boundary $\sqrt{x_1^2 + x_2^2} > \tau$ and *outperforms* gradient boosting ($F_1 = 0.996$ vs. 0.989). On **Breast Cancer** and **Ionosphere**, simple two-feature formulas achieve 95–98% of ensemble performance. On **Moons**, the nonlinear boundary resists depth-3 formulas (gap = 0.136). On the **Three-body** problem, the gap is largest (0.35), indicating substantial nonlinear structure.

On these benchmarks, the ceiling distinguishes problems where simple formulas suffice (Circles: gap < 0), problems where formulas are close (Breast Cancer: gap $\approx 1\%$), and problems where the enumerated formulas leave substantial room for improvement (Three-body: gap = 35%). All gaps use ensemble and ceiling on the same data and CV splits.

4 Case Study: Three-Body Periodic Orbit Prediction

4.1 Setup

We use a single dataset of 15,000 random configurations of equal-mass bodies ($m_1 = m_2 = m_3 = 1$). Both the formula search and the gradient-boosted baseline are evaluated on this dataset using the same 5-fold cross-validation splits, ensuring the ceiling and ensemble F1 are directly comparable.

Features include: numerical matrix rank (threshold $\epsilon = 10^{-6}$), potential block eigenvalues and their ratios, frequency cluster counts, spectral entropy, Hamiltonian energy, body separation ratios, and derived quantities. Ground truth (periodic vs. non-periodic) is obtained by symplectic leapfrog integration ($\Delta t = 0.005$, 8×10^4 steps, $T = 400$ natural time units) on an NVIDIA GV100 GPU, with periodicity defined as phase-space return to within 5% of initial extent after the first 10% of integration time. Batch size is maximized automatically using the batch-probe library [5].

4.2 Search Results

Table 2 summarizes the results across all four phases.

Table 2: Formula search results by phase (15,000 configs, 23 features). All phases search formulas of depth ≤ 3.

Phase (depth)	Best formula	F1	Tested
1: single+unary (1–2)	`n_clusters`	0.35	138
2: pairwise (2)	`n_clust +` $\lambda_{\min}$	0.48	3,542
3: unary of top-30 (3)	`hypot(n_cl,` $\lambda_{\min}$`)`	0.48	150
4: triple of top-8 (3)	(none above 0.30)	—	224
Best (depth ≤ 3)		**0.48**	**4,054**

The best F1 is reached at Phase 2 and does not improve in Phases 3–4. All 20 top formulas are algebraically equivalent to linear combinations of n_{clusters} and $\lambda_{\min}$. The Euclidean norm variant

$$\hat{y} = \mathbf{1}\left[\sqrt{n_{\text{clusters}}^2 + \lambda_{\min}^2} > \tau\right] \tag{1}$$

is representative of the equivalence class.

4.3 Irreducible Complexity Gap

Table 3: Formula ceiling vs. black-box models (all on 15K dataset, 5-fold CV for ensemble methods; $\pm$ shows fold standard deviation).

Model	F1	Gap
Random (4% base rate)	$\sim$0.07	—
$r_2/r_1 > 10$	0.29	—
Best enumerated formula	0.48	0.00
Decision tree (CV)	0.55 ± 0.03	-0.07
Gradient boosting (CV)	0.83 ± 0.02	-0.35

The observed gap is $G = 0.83 - 0.48 = 0.35$ (both measured on the same 15K dataset with 5-fold CV for the ensemble). The decision tree (F1 = 0.55, CV) captures some of this gap via piecewise-constant splits; the remaining $0.83 - 0.55 = 0.28$ requires the nonlinear interactions of an ensemble. Within the enumerated formula structures, no expression achieves more than 58% of the ensemble's predictive power on this dataset. This gap may narrow with deeper search, richer operations, or additional features, but its stability across Phases 2–3 and across all tested operation types is consistent with genuine data complexity, though a richer formula space (e.g., multi-term sums, deeper nesting) could in principle narrow the gap.

4.4 Physical Interpretation

The discovered formula combines two independent necessary conditions:

- $n_{\text{clusters}} \geq 2$: the eigenvalue frequency spectrum separates into fast (inner binary) and slow (outer body) groups, indicating timescale separation.
- $\lambda_{\min} > 0$: the smallest Hessian eigenvalue is nonzero, indicating the system is gravitationally bound (not escaping).

The Euclidean norm indicates these contribute independently—there is no synergy or interaction between them. Periodicity requires *both* separated timescales *and* sufficient binding.

4.5 Comparison to PySR

We compare directly against PySR [4] on four AI Feynman regression problems (Table 4).

Table 4: Regression: our method vs. PySR (R^2, 3,000 samples).

Problem	Ours	PySR	PySR time	PySR formula
Gravity	0.99	**1.00**	21s	`m1*m2/r^2` (exact)
Coulomb	0.99	**1.00**	5s	`q1*q2/r^2` (exact)
$x_1^2+x_2^2$	0.50	**1.00**	6s	`sq(x1)+sq(x2)` (exact)
Thin lens	0.93	0.96	7s	approximate

PySR recovers the exact formula on 3/4 problems by exploring deeper trees (depth 10+). Our method matches PySR on Gravity and Coulomb (where depth 2 suffices: $m_1 m_2/r^2$) but misses $x_1^2 + x_2^2$ (which requires two additive squared terms—a structure outside our Phase 2 template). Note that this comparison uses regression (R^2) rather than our method's native classification (F1) objective, so it tests formula-recovery ability rather than the ceiling framework directly. A classification-side comparison against PySR (adapting its output to a thresholded binary predictor) would provide a more direct test but is not yet implemented. The key distinction between the methods remains: PySR produces a single best formula at each complexity; our framework additionally measures the ceiling and gap.

4.6 Convergence and Ablation

Table 5 shows the ceiling at each search phase.

Table 5: Ceiling convergence: F1 at each phase.

Dataset	Phase 1	Phase 2	Phase 3
Circles	0.730	**0.996**	0.996
Moons	**0.833**	0.833	0.833
Breast Cancer	0.926	**0.934**	0.934

The ceiling converges at Phase 2 for all datasets. Phase 3 (unary transforms) does not improve the best score, suggesting that the ceiling has stabilized for the tested operations.

Ablation. On Circles, removing the `hypot` operation drops the ceiling from 0.996 to 0.796. All other operations are redundant—`hypot` alone captures the circular decision boundary. This demonstrates that the ablation identifies which operations are structurally necessary.

4.7 Runtime Scaling

Table 6: Runtime vs. feature count (Phase 1+2, single CPU core).

d	Formulas	Time (s)	Best F1
2	16	0.01	0.96
5	145	0.09	0.98
10	640	0.38	0.96
15	1,485	0.91	0.97
20	2,680	1.66	0.95

Runtime scales as $O(d^2)$, reaching 1.7 seconds at $d = 20$. GPU vectorization (not used here) would further reduce these times. At $d = 50$, the estimated Phase 2 time is ∼20 seconds on CPU.

5 Discussion

5.1 When to Use This Method

The phased exhaustive approach is most valuable when: (i) the feature count is moderate ($d < 50$, so d^2 is tractable); (ii) the practitioner needs the best formula within a bounded depth, not a stochastic approximation; (iii) the formula ceiling and irreducible complexity gap are of interest (e.g., to decide whether a problem admits a simple law or requires a complex model).

For large feature spaces ($d > 100$) or deep formula trees (depth > 5), genetic methods (PySR) or neural symbolic regression (AI Feynman) are more appropriate. Our method's advantage is the ceiling diagnostic: PySR finds better formulas on deep problems, but does not quantify when further search is futile.

5.2 The Ceiling as a Scientific Diagnostic

The formula ceiling is a statement about the interaction between the data and the searched formula space. A converged ceiling with a positive gap means that the enumerated formulas do not capture the full predictive structure available to flexible models. This could indicate genuine nonlinearity in the data, or it could indicate that the searched templates miss a formula that would close the gap. In our three-body case study, the ceiling of 0.48 is stable across Phases 2–3 and across all 7 binary and 5 unary operations tested, which is consistent with—but does not prove—genuine complexity beyond the depth-3 formula space.

This diagnostic is novel. Existing symbolic regression methods report the best formula found but do not quantify *how much better* a more complex model could do. The ceiling provides a principled stopping criterion: if $C_{D+1} = C_D$ and $G_D > 0$, stop searching for formulas and use an ensemble.

5.3 Limitations

The search is exhaustive only within each phase's structure. Phase 2 (pairwise) is complete; Phase 3 applies unary transforms only to top-K pairwise results, not all; Phase 4 tests only pre-specified triple patterns. A formula with structure not matching any phase template (e.g., $\log(x_1 + x_2) \cdot x_3/x_4$) will be missed.

The method does not discover features—it evaluates pre-computed ones. The quality of the discovered formula depends entirely on the feature engineering.

The F1 metric weights precision and recall equally. Domain-specific objectives (e.g., prioritizing recall for safety applications) could change the optimal formula.

6 Conclusion

We have introduced phased enumeration with formula ceiling detection for symbolic prediction. The method enumerates all formulas within defined structural templates and measures the complexity gap to a black-box ensemble on the same data split. Applied to the three-body problem, the ceiling stabilizes at F1 = 0.48 while gradient boosting reaches 0.83 on the same dataset, yielding a gap of 0.35. Across five classification benchmarks, the ceiling correctly identifies problems where formulas suffice (Circles: ceiling beats ensemble) and where they do not (three-body, Moons).

The method is domain-general, requiring only a feature matrix and binary labels. It is practical for feature spaces up to $d \approx 50$ and formula depths up to 3, with GPU acceleration enabling evaluation of ∼6,000 formulas in under one minute.

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